

[UPDATE] Get to know and Control big.LITTLE

Post Reply



90 posts

1

2



mdrjr
Site Admin



[UPDATE] Get to know and Control big.LITTLE

Tue Oct 22, 2013 10:22 am

Obligatory reading:

CODE: [SELECT ALL](#)

Q: What's a cluster

A: The Exynos 5410 SoC has two CPU Clusters:

- One CPU cluster has 4 ARMv7-a Cortex-A7 cores running at a max frequency of 1.2Ghz, those 4-cores share a 25
- One CPU cluster has 4 ARMv7-a Cortex-A15 cores running at a max frequency of 1.6Ghz(1.8Ghz Overclock), those

1. Identify what CPU Cluster you are using.

Its very simple..

grep "CPU part" /proc/cpuinfo

if CPU part is 0xc07 them you are on the Little cores.. the A7's

if CPU part is 0xc0f them you are on the big cores.. the A15's

Hope it clarify for people asking on what cores they are 😊

2. Read big.LITTLE status

Another easy thing to do!

cat /dev/bL_status

CODE: [SELECT ALL](#)

```

0 1 2 3 L2 CCI
[A15] 0 0 0 0 0
[A7] 0 0 0 1 1
```

You'll what cores of the 4-cores of each cluster is alive and where l2 is 1 its the alive cluster 😊

3. Control big.LITTLE

Modes are:

CODE: [SELECT ALL](#)

```

00 - disables the cluster switching
01 - LITTLE only (only a7 cores)
11 - enables big.LITTLE cluster migration
```

BIG NOTE: BEFORE Switching between big and LITTLE manually, PLEASE disable cluster switching otherwise your board will lock and requires a reboot.

BIG NOTE2: Make sure that you are on ondemand governor!!!!!!!!!!

echo ondemand > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor

control method..

echo value > /dev/b.L_operator

example: echo 00 > /dev/b.L_operator

Small note: There's an extra mode.. 10 - big only, however its ignored by code, so, if you want purely A15 cores, switch to performance governor!

4. Benchmark Tests

My Setup:

Processor: ARMv7 rev 2 @ 1.80GHz (4 Cores), **Motherboard:** ODROIDXU, **Memory:** 2048MB, **Disk:** 40GB DSA2CT040G3 + 128GB Ext Hard Disk + 2GB uSD

OS: Ubuntu 13.10, **Kernel:** 3.4.5 (armv7l), **Compiler:** GCC 4.8, **File-System:** ext4, **Screen Resolution:** 1280x800

Ubuntu 13.10 with an UNMODIFIED kernel from the built in kernels

Phoronix Test Suite C-Ray 1.1 (Lower is better)

- A7 Only

CODE: [SELECT ALL](#)

```
C-Ray 1.1:
pts/c-ray-1.1.0
Test 1 of 1
Estimated Trial Run Count:    3
Estimated Time To Completion: 14 Minutes
  Started Run 1 @ 06:18:03
  Started Run 2 @ 06:24:26
  Started Run 3 @ 06:30:47 [Std. Dev: 0.47%]

Test Results:
  380.059
  377.929
  376.511
```

- A15 only

CODE: [SELECT ALL](#)

```
pts/c-ray-1.1.0
Test 1 of 1
Estimated Trial Run Count:    3
Estimated Time To Completion: 17 Minutes
  Started Run 1 @ 06:42:02
  Started Run 2 @ 06:44:14
  Started Run 3 @ 06:46:22 [Std. Dev: 1.95%]

Test Results:
  128.708
  125.254
  123.953

Average: 125.97 Seconds
```

7-Zip Compression 9.20.1 (higher is better)

- A7 only

CODE: [SELECT ALL](#)

```
pts/compress-7zip-1.6.0
Test 1 of 1
Estimated Trial Run Count:    3
Estimated Time To Completion: 4 Minutes
    Started Run 1 @ 07:07:47
    Started Run 2 @ 07:11:36
    Started Run 3 @ 07:15:24 [Std. Dev: 1.91%]

Test Results:
    862
    866
    836

Average: 855 MIPS
```

- A15 only

CODE: [SELECT ALL](#)

```
pts/compress-7zip-1.6.0
Test 1 of 1
Estimated Trial Run Count:    3
Estimated Time To Completion: 4 Minutes
    Started Run 1 @ 06:59:13
    Started Run 2 @ 07:00:22
    Started Run 3 @ 07:01:32 [Std. Dev: 0.69%]

Test Results:
    3119
    3162
    3145

Average: 3142 MIPS
```

janrinze



Re: Identify if you are in A7 or A15's cores.

📅 Wed Oct 23, 2013 1:51 am

Thanks! that is really nice to know.



sert00



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Mon Oct 28, 2013 9:32 am

wow!great to see this!Now the only thing missing is xu..tracking say it had to be arrived on 25..but nothing.hpe fedex will bring it here tomorrow!it's 20 days i'm without it,and excited to try this thing soon with the lcd kit i ordered too! 😄
Question-also before,with performance governor,only a15 wrked..or it's something added nw in commits you inserted?I'd plan to use only big clusters..

2x Odroid-U2
2x Odroid-XU-E
Odroid-U3
Odroid-C1+
Odroid-XU4
Lcd touch panel 10.1"
....and many of the goodies available....

mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

Mon Oct 28, 2013 9:39 am

sert00 wrote:

wow!great to see this!Now the only thing missing is xu..tracking say it had to be arrived on 25..but nothing.hpe fedex will bring it here tomorrow!it's 20 days i'm without it,and excited to try this thing soon with the lcd kit i ordered too! 😊
Question-also before,with performance governor,only a15 wrked..or it's something added nw in commits you inserted?I'd plan to use only big clusters..

Its the standard behave of the performance governor 😞
It will move directly to the A15's



sert00



Re: [UPDATE] Get to know and Control big.LITTLE

Mon Oct 28, 2013 10:37 am

ok,thanks! 😊

2x Odroid-U2
2x Odroid-XU-E
Odroid-U3
Odroid-C1+
Odroid-XU4
Lcd touch panel 10.1"
....and many of the goodies available....

d-extremex



Re: [UPDATE] Get to know and Control big.LITTLE

Mon Oct 28, 2013 7:12 pm

newbie question.

to set to use purely A15 cores,

is this correct

```
echo 00 > /dev/b.L_operator  
echo performance > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor  
echo 10 > /dev/b.L_operator
```

and how to keep the setting after a reboot

cheers



sert00



Re: [UPDATE] Get to know and Control big.LITTLE

Mon Oct 28, 2013 8:42 pm

do a init.d script..

2x Odroid-U2
2x Odroid-XU-E
Odroid-U3
Odroid-C1+
Odroid-XU4
Lcd touch panel 10.1"
....and many of the goodies available....



mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

Tue Oct 29, 2013 12:08 am

“ d-extremex wrote:

newbie question.

to set to use purely A15 cores,

is this correct

```
echo 00 > /dev/b.L_operator  
echo performance > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor  
echo 10 > /dev/b.L_operator
```

and how to keep the setting after a reboot

cheers

That's not a noob question. Its a lazy person who didn't read the text question..
echo performance > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor

only this line is enough..



Goerlitzer



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Nov 01, 2013 1:06 am

Just see that the terminal program "nmon" (sudo apt-get install nmon) shows if you press "r" every two (or one if you press "-") second which cpu is used. It's a great simple tool to see what's going on on your machine (cpu, sd-/eMMC-cards, network etc.) without stressing the cpu.



memeka



Re: [UPDATE] Get to know and Control big.LITTLE

Tue Nov 12, 2013 10:32 am

From git odroidxu-next:

“

ARM: EXYNOS: add Exynos Dual Cluster Support ...
Add EDCS(Exynos Dual Cluster Support) for Samsung Exynos5410 SoC.
This enables all 8 cores, 4 x A7 and 4 x A15 run at the same time.

Is this working? 😊

Images: [U2/U3 Trusty Dev Center](#) | [XU Trusty Dev Center](#) | [XU4 Hipster Stretchy Pants](#)
Information: [U2/U3 Dashboard](#) | [XU Dashboard](#)
[Say thank you with a beer](#)

mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

Tue Nov 12, 2013 10:34 am

“ memeka wrote:

From git odroidxu-next:

“

ARM: EXYNOS: add Exynos Dual Cluster Support ...
Add EDCS(Exynos Dual Cluster Support) for Samsung Exynos5410 SoC.
This enables all 8 cores, 4 x A7 and 4 x A15 run at the same time.

Is this working? 😊

No and this is not the place to post this.



memeka



Re: [UPDATE] Get to know and Control big.LITTLE

Tue Nov 26, 2013 6:45 pm

just a small note:

the L2 is on (1) for the A7 cluster even if A15 is being used.

EDIT: I think this is why I constantly see A7 using 0.01-0.02W of power even though I am running on A15 (which use on idle more than 0.1W)

Images: [U2/U3 Trusty Dev Center](#) | [XU Trusty Dev Center](#) | [XU4 Hipster Stretchy Pants](#)
Information: [U2/U3 Dashboard](#) | [XU Dashboard](#)
[Say thank you with a beer](#)

usee



Re: [UPDATE] Get to know and Control big.LITTLE

Wed Nov 27, 2013 6:04 am

I am running performance tests with Java on a multithreaded application.

With default settings in debian the board does not deliver all performance but it changes frequency and switches cores on and off.

When I apply `echo performance > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor`
then the board uses all A15 cores at 1600 MHz.

Shall the board deliver 1800 MHz if I do the following:

`echo 1800000 > /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq?`

Is it safe to the above?

usee



Re: [UPDATE] Get to know and Control big.LITTLE

Wed Nov 27, 2013 8:11 am

Thanks @memeka, it is so one must first type 'sudo su' prior one can echo performance governor.

And it is not possible to set scaling_cur_freq, instead it shows what frequency the board is now using.

I overclocked the board to 1.8 GHz by setting the min to same value with max as follows:

```
#cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_max_freq > /sys/devices/system/cpu/cpu0/cpufreq/scaling_min_freq
```

Then the board gives the best performance so far in my test. I inspect the used frequency as follows:

```
#cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq  
1800000
```

I think that this is the max that I will test. The board behaves so that, when it is under load and gets hot it is swithes cores on and off while keeping the frequency. For instance

```
#cat /dev/bL_status
```

```
0 1 2 3 L2 CCI
```

```
[A15] 1 1 0 1 1
```

```
[A7] 0 0 0 0 0
```

So, I set the min frequency now back to original value.

Thanks, for the information that is given under this topic!



roosemberth



Re: [UPDATE] Get to know and Control big.LITTLE

Wed Nov 27, 2013 8:18 am

Is it possible to get to 2GHz with an excelent cooling system?, which are the barriers? just Thermal Throttling? or there are more limitators?



memeka



Re: [UPDATE] Get to know and Control big.LITTLE

Wed Nov 27, 2013 8:47 am

usee wrote:

Thanks @memeka, it is so one must first type 'sudo su' prior one can echo performance governor.

And it is not possible to set scaling_cur_freq, instead it shows what frequency the board is now using.

I overclocked the board to 1.8 GHz by setting the min to same value with max as follows:

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#cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_max_freq > /sys/devices/system/cpu/cpu0/cpufreq/scaling_min_freq
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Then the board gives the best performance so far in my test. I inspect the used frequency as follows:

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#cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq  
1800000
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I think that this is the max that I will test. The board behaves so that, when it is under load and gets hot it is swithes cores on and off while keeping the frequency. For instance

```
#cat /dev/bL_status
```

```
0 1 2 3 L2 CCI
```

```
[A15] 1 1 0 1 1
```

```
[A7] 0 0 0 0 0
```

So, I set the min frequency now back to original value.

Thanks, for the information that is given under this topic!

"sudo su" ? 😄 that's funny

you can just do "su" and enter root password
or you can enter "sudo bash" and enter user password.

CODE: [SELECT ALL](#)

```
root@elhokar ~ # echo performance > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
root@elhokar ~ # cat /dev/bL_status
      0 1 2 3 L2 CCI
[A15]  1 1 0 1  1
[A7]   0 0 0 0  1
```

You can see here I don't have "0 0 0 0" for A7, the L2 cache is still on.

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Information: [U2/U3 Dashboard](#) | [XU Dashboard](#)
[Say thank you with a beer](#)

usee



Re: [UPDATE] Get to know and Control big.LITTLE

Thu Nov 28, 2013 3:44 am

Please, just don't mind how I was writing about sudo and su. I am using different distros and in them behaviour may vary. I am using Fedora at the moment. In the internet there is enough discussion on sudo and su.

Anyway, the performance tuning is a rich topic. I just need to know enough of it so that I can inspect that different distros have same parameters. So, I rather leave explanations to be written by such people who know how the frequency management is designed to work and how it is aimed to be tuned. Thanks.

lizardmech



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Dec 06, 2013 9:13 pm

roosemberth wrote:

Is it possible to get to 2GHz with an excelent cooling system?, which are the barriers? just Thermal Throttling? or there are more limitators?

It appears to have the same voltage as 1.8ghz in the kernel files I found, I can't work out how to set it to 2ghz though, trying to set 2ghz through the kernel or /cpuo/cpufreq/ was ignored. I was able to make a custom kernel that used 1.8ghz on all governors without any issues.

grsingla



Re: [UPDATE] Get to know and Control big.LITTLE

Thu Jan 02, 2014 2:23 am

Hi,

Thanks everyone. I have a odroid XU+E and do the same thing.

I want to run an application only on A7 core and then only on A15 core and compare their performance.

I am not aware of the toolchain as to which software to be used, which language and how to interface it with the board.

Which OS should i use for it?

Can you please guide me through it?



odroid
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Thu Jan 02, 2014 12:29 pm

“ lizzardmech wrote:

It appears to have the same voltage as 1.8ghz in the kernel files I found, I can't work out how to set it to 2ghz though, trying to set 2ghz through the kernel or /cpuo/cpufreq/ was ignored. I was able to make a custom kernel that used 1.8ghz on all governors without any issues.

Did you guys run XU at 1.8Ghz?

According to our various tests, it was really unstable at 1.8Ghz.



odroid
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Thu Jan 02, 2014 12:30 pm

“ grsingla wrote:

Hi,

Thanks everyone. I have a odroid XU+E and do the same thing.

I want to run an application only on A7 core and then only on A15 core and compare their performance.

I am not aware of the toolchain as to which software to be used, which language and how to interface it with the board.

Which OS should i use for it?

Can you please guide me through it?

Please make your own thread for asking.

BTW, you can use any OS image with the latest Kernel update.



grsingla



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Fri Jan 03, 2014 4:55 am

Hi,

Thanks for helping on how to operate big cores and little cores separately.

I wanted to know are we able to operate the cores within a cluster separately or it supports cluster wise migration only.

Also can we only use IKS or global task scheduling (GTS) is also supported.

And we can operate all the 8 cores at the same time right?



mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Fri Jan 03, 2014 5:11 am

GTS isn't possible on Exynos 5410.

Just cluster migration works.



grsingla



Re: [UPDATE] Get to know and Control big.LITTLE

 Fri Jan 03, 2014 5:30 am

Hi,

Thanks and all the 8 cores can be operated at the same time other than being operated separately right?



mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

 Fri Jan 03, 2014 9:45 am

“ grsingla wrote:

Hi,

Thanks and all the 8 cores can be operated at the same time other than being operated separately right?

Nope, it can't.



grsingla



Re: [UPDATE] Get to know and Control big.LITTLE

 Fri Jan 03, 2014 9:59 am

Hi,

I have android 4.2.2 on odroid XU+E.

after i write command

cat /dev/bL_status

it gives me an error permission denied.

Can you tell me how to get rid of this error.



mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

 Fri Jan 03, 2014 10:06 am

“ grsingla wrote:

Hi,

I have android 4.2.2 on odroid XU+E.

after i write command

cat /dev/bL_status

it gives me an error permission denied.

Can you tell me how to get rid of this error.

Run it as root? type su before 😊



grsingla



Re: [UPDATE] Get to know and Control big.LITTLE

 Fri Jan 03, 2014 10:54 am

Hi,

Thanks..wen i run only A15 the freq is constant 1600Mhz..does stopping cpu migration and manually controlling the migration using

governor turns off the DVFS.

mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Jan 03, 2014 11:02 am

Yes.. that's right

grsingla



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Jan 03, 2014 11:09 am

thanks..wen i run A15 freq is 1600Mhz but wen i migrate to A7 manually i was expecting it to show 1200Mhz but still it was showing 1600Mhz.

Can you help me with this?



sert00



Re: [UPDATE] Get to know and Control big.LITTLE

Sat Jan 04, 2014 10:50 pm

Why don't you open a 3ad where you ask for help,like odroid and mdrjr already sad you?
and why ask via PM to me,to mdrjr etc..when you already asked the same thing in 2 different 3ads?don't go off-topic everywhere,open a 3ad

2x Odroid-U2
2x Odroid-XU-E
Odroid-U3
Odroid-C1+
Odroid-XU4
Lcd touch panel 10.1"
....and many of the goodies available....

lizardmech



Re: [UPDATE] Get to know and Control big.LITTLE

Sun Jan 05, 2014 11:29 pm

“ odroid wrote:

“ lizardmech wrote:

It appears to have the same voltage as 1.8ghz in the kernel files I found, I can't work out how to set it to 2ghz though, trying to set 2ghz through the kernel or /cpuo/cpufreq/ was ignored. I was able to make a custom kernel that used 1.8ghz on all governors without any issues.

Did you guys run XU at 1.8Ghz?

According to our various tests, it was really unstable at 1.8Ghz.

I never have any trouble at 1.8ghz, I use it as my everyday desktop PC and get weeks of uptime. Maybe exynos5 SoCs vary a lot between production weeks?

dezza



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 2:36 am

What ubuntu version are you running? Something custom?
I'm getting Permission denied with root , 13.10 but got apt-get upgrade earlier.

```
root@odroid:/home/odroid# echo 1800000 > /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq
bash: /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq: Permission denied
```

[quote="mdrjr"]Obligatory reading:

CODE: [SELECT ALL](#)

```
Q: What's a cluster
A: The Exynos 5410 SoC has two CPU Clusters:
    - One CPU cluster has 4 ARMv7-a Cortex-A7 cores running at a max frequency of 1.2Ghz, those 4-cores share a 25
    - One CPU cluster has 4 ARMv7-a Cortex-A15 cores running at a max frequency of 1.6Ghz(1.8Ghz Overclock), those
```



mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 2:44 am

scaling_cur_freq is readonly!
scaling_max_freq and scaling_min_freq are rw



dezza



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 3:09 am

Of course! Thanks.
Back to 13.10 then. 12.04 allowed to set max, but was keeping it at 1.6

```
root@odroid-desktop:/home/odroid# echo 1800000 > /sys/devices/system/cpu/cpu0/cpufreq/scaling_max_freq
root@odroid-desktop:/home/odroid# cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_max_freq
1600000
```



mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 3:14 am

1.8Ghz only works on ondemand.



dezza



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 3:24 am

Still no love 🙄

```
root@odroid:/sys/devices/system/cpu/cpu0/cpufreq# echo ondemand > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
```

```
root@odroid:/sys/devices/system/cpu/cpu0/cpufreq# echo 1800000 > /sys/devices/system/cpu/cpu0/cpufreq/scaling_max_freq
root@odroid:/sys/devices/system/cpu/cpu0/cpufreq# cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_max_freq
1600000
root@odroid:/sys/devices/system/cpu/cpu0/cpufreq# cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
ondemand
root@odroid:/sys/devices/system/cpu/cpu0/cpufreq#
```

dezza



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 8:01 am

Both 13.10 and 12.04 did not set max at 1.8, and stuck at 1.6GHz in either ondemand or performance mode.
Is there known version number was capable of running at 1.8GHz ?
Thanks!

Last edited by dezza on Fri Feb 07, 2014 10:08 am, edited 1 time in total.



memeka



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 9:02 am

if you compile your own kernel you can set up frequency there at up to 2GHz.

Images: [U2/U3 Trusty Dev Center](#) | [XU Trusty Dev Center](#) | [XU4 Hipster Stretchy Pants](#)

Information: [U2/U3 Dashboard](#) | [XU Dashboard](#)

[Say thank you with a beer](#)

roosemberth



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 9:26 am

You should look up for a more efficient cooling solution, Thermal Thottling will cause the processor frequency to drop (until it reaches -82°C) when it reaches 90°C-95°C, this is software triggered (See HOT_NORMAL_TEMP and HOT_CRITICAL_TEMP in drivers/thermal/exynos_thermal.c). If it gets above 100°C, It'll drop the freq to 200MHz (AFAIK), and @123°C Hardware-Triggered will power it off if reached.

I was thinking in a water cooling, but the Idea of a leakage freaks me out, I might take the risk through...

Best Regards!

dezza



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 10:07 am

@memeka, I thought this thread was implying stock Ubuntu can be overclocked. Any pointers on where to look for 2GHz unlock for recompile highly appreciated

@roosemberth, thank, just benchmarking now to see how it will stack against Snap Dragon at similar frequency.

mdrjr

Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 9:02 pm

Gimme a couple of days and I'll check that for you.. not at home ATM



odroid
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

Fri Feb 07, 2014 11:09 pm

When we tested the over-clocking with 10 XU boards, only 2 boards can run at 1.8Ghz.
Other 8 boards' maximum frequency was only 1.7 or 1.6Ghz.

So 2.0 Ghz over-clocking seems to be almost impossible or extremely rare case.



memeka



Re: [UPDATE] Get to know and Control big.LITTLE

Sat Feb 08, 2014 6:07 am

oh I'm sorry I was talking about the U2 😊

Images: [U2/U3 Trusty Dev Center](#) | [XU Trusty Dev Center](#) | [XU4 Hipster Stretchy Pants](#)
Information: [U2/U3 Dashboard](#) | [XU Dashboard](#)
[Say thank you with a beer](#)

Bohemund



Re: [UPDATE] Get to know and Control big.LITTLE

Mon Feb 10, 2014 9:12 pm

Hello,

I get a "bash: /dev/b.L_operator: Permission denied
" on all commands if i execute them with sudo - whats my mistake ? 😊

“ d-extremex wrote:

newbie question.

to set to use purely A15 cores,

is this correct

```
echo 00 > /dev/b.L_operator  
echo performance > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor  
echo 10 > /dev/b.L_operator
```

and how to keep the setting after a reboot

cheers

mdrjr
Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Mon Feb 10, 2014 9:16 pm

You sure that you are running as root ?



dezza



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Tue Feb 11, 2014 3:07 am

mdrjr,
Any luck checking "overclockable" kernel version/availability? I've tried recompile to 3.4.76 - it is still locked to 1.6Ghz. Not quite shire which part unlocks it .

It seems to be checking in multiple places in code so not very obvious. There some legacy Inkernel code as well.

Looks like 3.8, 3.11. 3.12 kernels are not XU compatible yet, no luck booting with it.

“ mdrjr wrote:

Gimme a couple of days and I'll check that for you.. not at home ATM



roosemberth



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Tue Feb 11, 2014 3:50 am

I've seen on github the cpu-freq code: [https://github.com/hardkernel/linux/blo ... -cpufreq.c](https://github.com/hardkernel/linux/blob/master/arch/arm/mach-exynos/driver/cpufreq/cpufreq.c) But I can't find where the implementation is...

I was told to check in governors and related, but, I hadn't take a look yet

Best Regards!



dezza



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Tue Feb 11, 2014 3:59 am

I think those are for inkernel old stuff?

Seem that new is here, and some other files in the folder, really spread out accross the files

[https://github.com/hardkernel/linux/blo ... ynos5410.c](https://github.com/hardkernel/linux/blob/master/arch/arm/mach-exynos/driver/cpufreq/cpufreq.c)



mdrjr

Site Admin



Re: [UPDATE] Get to know and Control big.LITTLE

📅 Tue Feb 11, 2014 5:50 am

Nope, those instructions still work for XU.. I've just tested.

@roosemberth,

driver/cpufreq/*exynos*.c



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